# BLOCKCHAIN TECHNOLOGY

**(Practical 3)**

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**Problem Statement:** Write a smart contract on a test network, for Bank account of a customer for following operations: Deposit money, Withdraw Money, and Check Balance.

## SOURCE CODE

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract Bank {
    mapping(address => uint256) private balances;

    event Deposited(address indexed account, uint256 amount);
    event Withdrawn(address indexed account, uint256 amount);

    function deposit() public payable {
        require(msg.value > 0, "Deposit amount must be greater than 0");
        balances[msg.sender] += msg.value;
        emit Deposited(msg.sender, msg.value);
    }

    function withdraw(uint256 amount) public {
        require(amount > 0, "Withdraw amount must be greater than 0");
        require(balances[msg.sender] >= amount, "Insufficient balance");
        balances[msg.sender] -= amount;
        payable(msg.sender).transfer(amount);
        emit Withdrawn(msg.sender, amount);
    }

    function getBalance() public view returns (uint256) {
        return balances[msg.sender];
    }
}
```

# OUTPUT

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract Bank {
    // Mapping to store balances of each customer (a
    mapping(address => uint256) private balances;

    // Events for logging
    event Deposited(address indexed account, uint256 amount);
    event Withdrawn(address indexed account, uint256 amount);

    // Deposit money into the bank
    function deposit() public payable {
        require(msg.value > 0, "Deposit amount must be greater than 0");
        balances[msg.sender] += msg.value;
        emit Deposited(msg.sender, msg.value);
    }

    // Withdraw money from the bank
    function withdraw(uint256 amount) public {
        require(amount > 0, "Withdraw amount must be greater than 0");
```

uint256

contracts/Bank.sol 8:45

infinite gas

infinite gas

**Fig 1: Smart Contract Code Compilation in Remix IDE (Bank.sol)**

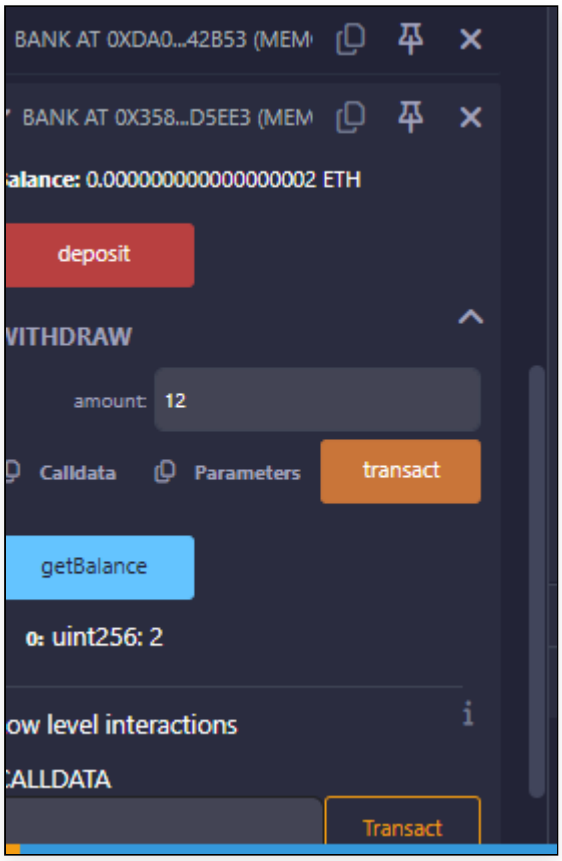



**Fig 2: Deployed Bank Contract with deposit, withdraw, and getBalance Functions**

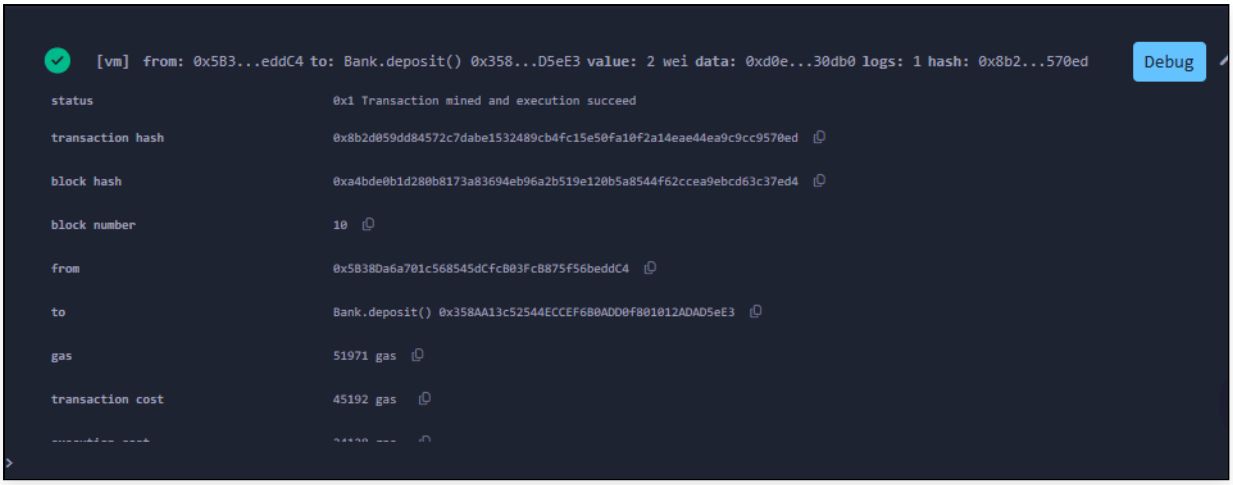



**Fig 3: Transaction Mined Receipt for deposit() with Gas Execution Details**

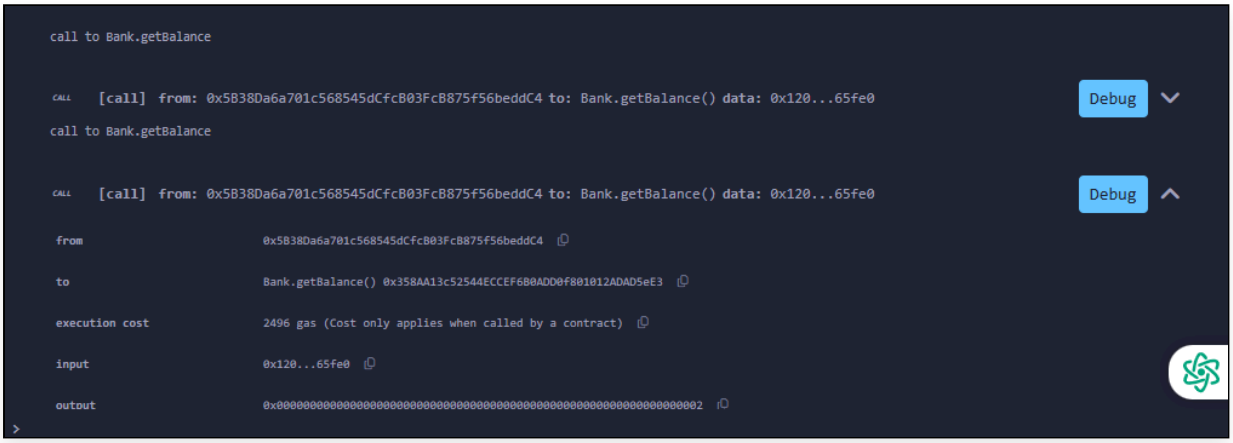



**Fig 4: Execution Receipt for getBalance() showing Updated Account Balance**